Reg. No. | | | | | | | | | | |

[18ESCS02]

THIRD SEMESTER B. Tech EXAMINATION, DECEMBER–2021

DIGITAL & LOGIC DESIGN

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Draw and Explain the logic gates and truth table for the following: **10**

- **i.** AND
- **ii.** OR
- **iii.** EX-OR
- **iv.** EX-NOR

b. State and prove DeMorgan's theorem. **04**

2. a. The Boolean expression Y = A (B + C D), draw logic gate and truth table. **09**

b. Define the following: **05**

- **i.** Maxterm
- **ii.** Literal
- **iii.** Minterm
- **iv.** SOP
- **v.** POS

3. a. Design the following circuits using gates **10**

- **i.** Half Adder circuit
- **ii.** Full Adder circuit

b. Define multiplexer with its advantages. **04**

OR

4. a. Design the following Circuits using gates **10**

- **i.** Half Subtractor Circuit
- **ii.** Full Subtractor Circuit

b. Define Demultiplexer with its advantages **04**

5. a. Obtain the characteristic equations and Excitation table for the following flip-flops. **10**

- **i.** SR flip-flop
- **ii.** JK flip-flop

	b.	Explain D flip flop with neat timing diagram.	**04**
		OR	
6.	**a.**	Explain 3 bit Asynchronous down Counter with neat diagram and wave forms.	**10**
	b.	Differentiate between Synchronous and Asynchronous circuits.	**04**
7.	**a.**	Illustrate working principle of CMOS INVERTER.	**10**
	b.	Application of Shift Registers.	**04**
		OR	
8.	**a.**	Explain CMOS circuit operation using NAND GATE?	**10**
	b.	Define Characteristics of digital IC's.	**04**